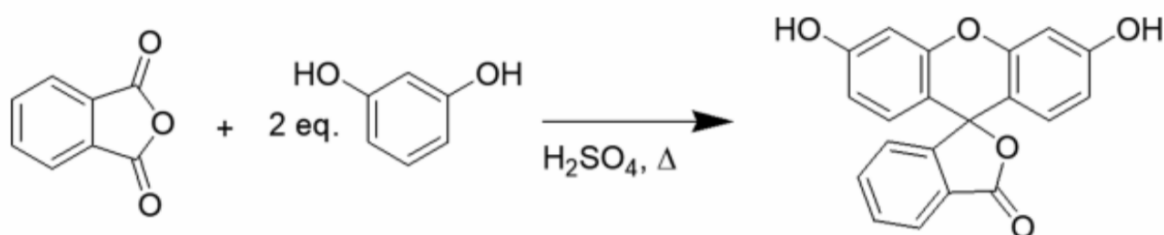


Experiment: Production of a Biphasic Glow Stick through Organic Synthesis, Fluorescence, and Chemiluminescence

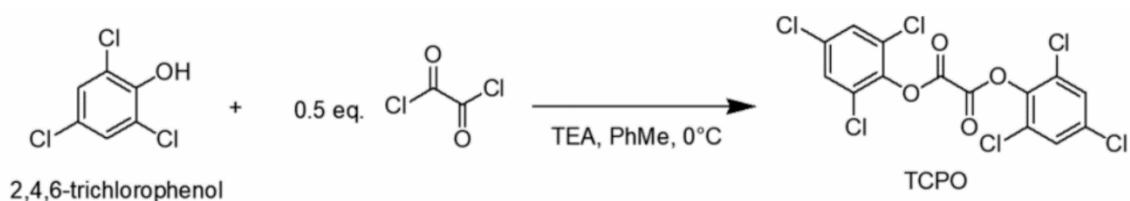
Procedure:

Step 1: Synthesis of fluorescein:



1. Crush 1gm (1 equivalent) of **Phthalic anhydride** and 1.4gm (2 equivalent) of **Resorcinol** using a mortar and pestle. Transfer the resulting powder and a magnetic stirrer bead into a 100 mL beaker.
2. Place the beaker in an oil bath on a magnetic stirrer. Heat the mixture to 150°C - 180°C until it melts.
3. Once melted, carefully add concentrated **Sulfuric** acid to the beaker.
4. Observe the colour change. Once the reaction occurs, remove the beaker from the stirrer and add more concentrated sulfuric acid to dissolve the thick paste.
5. After the paste has melted, add water to the beaker to dilute the mixture.
6. Vacuum filter the solution. The product, **fluorescein**, will be obtained as a yellowish powder.

Step 2: Synthesis of TCPO:



1. Place a 20 mL Test tube on a stirrer with a magnetic bead inside. Add 500mg (1 equivalent) of **2,4,6-trichlorophenol**, cover with a septum, and purge with nitrogen gas.
2. Dissolve the trichlorophenol in **Dry Toluene** using a metallic needle. Add 0.35mL (1 equivalent) of **Triethylamine** and stir for 5 minutes under nitrogen conditions.
3. After stirring, add 0.1mL (0.5 equivalent) of **Oxalyl chloride** to the solution while in an ice bath. The solution will solidify.
4. Remove the flask from the stirrer, vacuum filter the solution, and wash the solid with **Methanol**. The product, **TCPO**, will be a white powder.

Step 3: Glow stick assembly

1. **Fluorescein Solution:** Dissolve a small amount of **fluorescein powder** in a Sodium hydroxide solution to produce a green-fluorescent colour.
2. **Chemiluminescent Solution:** Prepare a solution of **Sodium acetate (25mg)** and **Diphenyl anthracene (25mg)** (1:1) in Diethyl phthalate (2ml). Add 5 drops of Hydrogen peroxide to the solution. Add a small amount of synthesized **TCPO** to the chemiluminescent solution. A blue chemiluminescent glow will be observed.
3. **Glow Stick:** Combine the fluorescent and chemiluminescent solutions in a test tube. Observe the glow stick effect in a dark room, where chemiluminescence and fluorescence will be visible.